

Canalicular method rebuilt around the physical scale

PIXEL SIZE 0.13 um/px. PIXEL SIZE 0.13 UM/PX, ROUNDED, UNCONFIRMED.
PRE-VALIDATION. No variant is called better than another.

Variants:
A current method, as is
P1 ridge, sigmas 0.13, 0.17, 0.21 um
P2 ridge, sigmas 0.17, 0.21, 0.26 um (primary)
P3 ridge, one scale per image chosen from the image
P4 P2 with the low cut at 0.9 of the high cut
P5 P2 with gap bridging off
N P2 on the notch filtered image

Mean over the 8 images

measure	A	P1	P2	P3	P4	P5	N
width_fwhm_median_um	0.6368	0.6314	0.6352	0.6409	0.6362	0.6351	0.6352
width_dt_median_um	0.7853	0.5353	0.6006	0.7465	0.5814	0.6006	0.6006
low_contrast_fraction	0.02063	0.05829	0.04053	0.03504	0.02528	0.04019	0.04044
contrast_snr_median	9.955	7.299	7.877	8.756	7.724	7.882	7.877
straight_run_fraction	0.221	0.2267	0.2375	0.247	0.2318	0.2365	0.2376
straight_run_fraction_smoothed	0.1872	0.1797	0.194	0.2038	0.1921	0.1931	0.1939
rectangle_count	0	0.25	0	0	0	0	0
connected_to_lacuna_fraction	0.336	0.4655	0.48	0.5373	0.4341	0.4557	0.48
junction_count	529.8	791.2	738.8	705.9	608.6	733.6	738.1
thread_end_fraction	0.7641	0.6959	0.6869	0.6671	0.716	0.6927	0.6871
field_length_density_um_per_um2	0.3008	0.3358	0.3237	0.3106	0.3029	0.3232	0.3236
roots_per_cell	7.545	8.279	7.965	7.508	7.483	7.965	7.965
skeleton_px	3.992e+04	4.456e+04	4.295e+04	4.122e+04	4.019e+04	4.289e+04	4.294e+04
n_bridges	18.12	21.25	19.62	21	85.12	0	19.38
legacy_unsupported_fraction	0.06798	0.1018	0.09214	0.08723	0.06008	0.09118	0.09202

Change from A, per cent

measure	A	P1	P2	P3	P4	P5	N
width_fwhm_median_um	+0.0	-0.9	-0.3	+0.6	-0.1	-0.3	-0.3
width_dt_median_um	+0.0	-31.8	-23.5	-4.9	-26.0	-23.5	-23.5
low_contrast_fraction	+0.0	+182.6	+96.5	+69.9	+22.5	+94.8	+96.1
contrast_snr_median	+0.0	-26.7	-20.9	-12.0	-22.4	-20.8	-20.9
straight_run_fraction	+0.0	+2.6	+7.5	+11.8	+4.9	+7.0	+7.5
straight_run_fraction_smoothed	+0.0	-4.0	+3.6	+8.9	+2.6	+3.2	+3.6
rectangle_count							
connected_to_lacuna_fraction	+0.0	+38.5	+42.9	+59.9	+29.2	+35.6	+42.8
junction_count	+0.0	+49.4	+39.5	+33.2	+14.9	+38.5	+39.3
thread_end_fraction	+0.0	-8.9	-10.1	-12.7	-6.3	-9.4	-10.1
field_length_density_um_per_um2	+0.0	+11.6	+7.6	+3.3	+0.7	+7.5	+7.6
roots_per_cell	+0.0	+9.7	+5.6	-0.5	-0.8	+5.6	+5.6
skeleton_px	+0.0	+11.6	+7.6	+3.3	+0.7	+7.5	+7.6
n_bridges	+0.0	+17.2	+8.3	+15.9	+369.7	-100.0	+6.9
legacy_unsupported_fraction	+0.0	+49.7	+35.5	+28.3	-11.6	+34.1	+35.4

pixel size 0.13 um/px, rounded, unconfirmed.

Pre-validation: no number here has been checked against a manual count.

A diagnosis and a better founded experiment. No variant is called better than another.

width_fwhm *: the full width at half maximum of the intensity profile across the thread, taken on the flattened image along the normal at each usable skeleton pixel. It INCLUDES the optical blur, so it is an apparent width in the image and an upper bound on the diameter, not the diameter. No point spread function is computed or subtracted: the numerical aperture, the emission wavelength and the pinhole are unknown.

width_dt *: the older measure, twice the distance transform of the mask, kept beside it. It can only return the width of the mask it is given.

contrast_snr: the intensity at a skeleton pixel minus the median intensity 4 to 6 px away along the normal on both sides, over the noise, where the noise is 1.4826 times the median absolute deviation of the flattened image outside the lacunae, the vascular regions and that variant's own mask. low_contrast_fraction is the share below 2.

legacy_unsupported_fraction: the earlier measure, which used the CURRENT method's own threshold and so favoured the current method. Kept only so the earlier tables can be followed.

straight_run_fraction: share of skeleton pixels in an axis-aligned run of at least 8 px. A thin skeleton on a pixel grid contains straight runs by its nature, and the current method already scores about 0.22, so only differences between variants mean anything here.

straight_run_fraction_smoothed: the same after each branch path is smoothed with a 5 point moving average, which removes most of the pixel grid effect.

Every image and variant

image	v	width_fwhm_median_um	width_dt_median_um	low_contrast_fraction	straight_run_fraction	straight_run_fraction_smoothed	rectangle_count	connected_to_lacuna_fraction	junction_count	field_length_density_um_per_um2	roots_per_cell	skeleton_px	n_bridges
542_z06	A	0.6512	0.8222	0.0091	0.2298	0.1958	0	0.3743	375	0.2665	8.615	3.496e+04	17
542_z06	P1	0.6484	0.5814	0.014	0.2245	0.1849	0	0.4353	529	0.2946	9.154	3.865e+04	14
542_z06	P2	0.6502	0.7354	0.0107	0.234	0.1972	0	0.437	494	0.287	9.385	3.764e+04	9
542_z06	P3	0.6542	0.78	0.016	0.2359	0.1942	0	0.4486	506	0.2819	8.769	3.698e+04	12

542_z06	P4	0.6497	0.5814	0.0076	0.2293	0.1942	0	0.4156	429	0.2722	9.077	3.571e+04	67
542_z06	P5	0.6502	0.7354	0.0106	0.2332	0.1964	0	0.4347	492	0.2866	9.385	3.76e+04	0
542_z06	N	0.6502	0.7354	0.0107	0.234	0.197	0	0.437	494	0.2869	9.385	3.764e+04	9
542_z18	A	0.645	0.78	0.0077	0.2141	0.1869	0	0.2993	395	0.2621	8.273	3.432e+04	14
542_z18	P1	0.6418	0.5814	0.015	0.2225	0.1823	0	0.3288	583	0.2917	8.909	3.82e+04	22
542_z18	P2	0.6447	0.5814	0.0112	0.2258	0.187	0	0.3082	566	0.2845	8.454	3.726e+04	10
542_z18	P3	0.6492	0.78	0.017	0.2493	0.2055	0	0.4004	572	0.2793	8.364	3.658e+04	16
542_z18	P4	0.6444	0.5814	0.0077	0.2222	0.1897	0	0.2969	474	0.2691	8.364	3.524e+04	71
542_z18	P5	0.6447	0.5814	0.011	0.2248	0.1864	0	0.3027	563	0.2843	8.454	3.723e+04	0
542_z18	N	0.6447	0.5814	0.0111	0.2259	0.1869	0	0.3081	564	0.2845	8.454	3.725e+04	10
543_z2	A	0.6307	0.78	0.0355	0.206	0.1669	0	0.4079	681	0.3439	7.583	4.578e+04	21
543_z2	P1	0.6236	0.52	0.1016	0.2143	0.1627	0	0.5379	990	0.3792	8.5	5.049e+04	24
543_z2	P2	0.6291	0.5814	0.0631	0.2289	0.1855	0	0.5402	866	0.355	7.833	4.726e+04	19
543_z2	P3	0.6366	0.7354	0.0518	0.2372	0.1934	0	0.6351	810	0.34	7.083	4.526e+04	23
543_z2	P4	0.6311	0.5814	0.0405	0.221	0.1828	0	0.4764	683	0.3312	7.083	4.41e+04	84
543_z2	P5	0.6291	0.5814	0.0627	0.2282	0.1849	0	0.5171	861	0.3545	7.833	4.721e+04	0
543_z2	N	0.6291	0.5814	0.0629	0.2288	0.1854	0	0.5399	866	0.3548	7.833	4.725e+04	18
543_3	A	0.6312	0.78	0.0406	0.1997	0.1662	0	0.2573	698	0.3472	6.889	4.642e+04	14
543_3	P1	0.6243	0.52	0.1107	0.2108	0.1621	0	0.3576	1044	0.3848	7.667	5.144e+04	26
543_3	P2	0.6295	0.5814	0.0796	0.2215	0.175	0	0.378	952	0.368	7.333	4.919e+04	29
543_3	P3	0.6379	0.7354	0.0597	0.2343	0.1913	0	0.5086	867	0.3465	6.778	4.633e+04	24
543_3	P4	0.6317	0.5814	0.0515	0.2176	0.176	0	0.2889	768	0.3408	6.667	4.556e+04	89
543_3	P5	0.6295	0.5814	0.0792	0.221	0.1748	0	0.3561	946	0.3673	7.333	4.91e+04	0
543_3	N	0.6295	0.5814	0.0796	0.2216	0.1751	0	0.3779	950	0.3679	7.333	4.918e+04	29
543_z13	A	0.6321	0.78	0.0338	0.2048	0.1744	0	0.3864	643	0.3369	6.727	4.503e+04	19
543_z13	P1	0.6256	0.52	0.0906	0.2115	0.1636	1	0.5349	939	0.3674	8.091	4.91e+04	21
543_z13	P2	0.6299	0.5814	0.063	0.2255	0.1831	0	0.6085	863	0.3512	7.273	4.694e+04	27
543_z13	P3	0.6363	0.7354	0.0481	0.2436	0.203	0	0.5581	780	0.333	6.818	4.451e+04	19
543_z13	P4	0.6315	0.5814	0.0385	0.2193	0.1808	0	0.5077	697	0.3268	6.909	4.368e+04	91
543_z13	P5	0.6299	0.5814	0.0627	0.2245	0.1822	0	0.5042	857	0.3506	7.273	4.686e+04	0
543_z13	N	0.6299	0.5814	0.0629	0.2256	0.1834	0	0.6084	864	0.3512	7.273	4.694e+04	27
682_z08	A	0.6349	0.78	0.0138	0.2446	0.2095	0	0.4099	480	0.2665	10	3.527e+04	24
682_z08	P1	0.6289	0.52	0.0414	0.2433	0.1949	0	0.5464	714	0.3052	11	4.039e+04	23
682_z08	P2	0.6325	0.5814	0.0306	0.259	0.2118	0	0.5679	680	0.2975	10.62	3.936e+04	20
682_z08	P3	0.6381	0.7354	0.0292	0.2624	0.2244	0	0.651	674	0.2856	10.25	3.779e+04	34
682_z08	P4	0.6336	0.5814	0.0183	0.2553	0.2088	0	0.5369	585	0.2778	10.12	3.676e+04	91
682_z08	P5	0.6325	0.5814	0.0303	0.2579	0.2108	0	0.556	677	0.2971	10.62	3.931e+04	0
682_z08	N	0.6325	0.5814	0.0305	0.2591	0.2116	0	0.5679	679	0.2974	10.62	3.936e+04	20
682_z23	A	0.6362	0.78	0.0134	0.2285	0.1921	0	0.3067	481	0.2946	6.727	3.917e+04	19
682_z23	P1	0.6311	0.52	0.0478	0.237	0.1885	0	0.5249	763	0.3335	6.727	4.434e+04	25
682_z23	P2	0.6342	0.5814	0.0355	0.2429	0.1939	0	0.554	767	0.3268	6.818	4.345e+04	24
682_z23	P3	0.639	0.7354	0.0334	0.2496	0.1997	0	0.6496	758	0.3167	6.545	4.211e+04	23
682_z23	P4	0.6352	0.5814	0.0211	0.2356	0.1885	0	0.5177	621	0.3057	6.364	4.064e+04	95
682_z23	P5	0.6341	0.5814	0.035	0.2404	0.1918	0	0.5406	760	0.3262	6.818	4.336e+04	0
682_z23	N	0.6342	0.5814	0.0355	0.2429	0.1938	0	0.5541	766	0.3267	6.818	4.344e+04	24
682_z29	A	0.6333	0.78	0.0111	0.2403	0.2055	0	0.2464	485	0.2883	5.545	3.837e+04	17
682_z29	P1	0.6274	0.52	0.0452	0.2494	0.1988	1	0.4581	768	0.3299	6.182	4.389e+04	15
682_z29	P2	0.6311	0.5814	0.0305	0.2625	0.2181	0	0.4463	722	0.3195	6	4.251e+04	19
682_z29	P3	0.6363	0.7354	0.0251	0.2641	0.2191	0	0.4469	680	0.3021	5.455	4.02e+04	17
682_z29	P4	0.632	0.5814	0.017	0.2541	0.2157	0	0.4327	612	0.2994	5.273	3.984e+04	93
682_z29	P5	0.6311	0.5814	0.03	0.2618	0.2176	0	0.434	713	0.319	6	4.245e+04	0
682_z29	N	0.6311	0.5814	0.0303	0.2627	0.2181	0	0.4464	722	0.3194	6	4.25e+04	18